INSTRUCTOR GUIDE

RLUSD Stablecoin Payments for Institutions

Module 1: Introduction to RLUSD and Institutional Payment Infrastructure

RUSD-001

1-day course · 5 modules · intermediate level

Learning Objectives

After completing this module, you will be able to:

- Describe the fundamental characteristics and regulatory foundation of RLUSD
- Explain how RLUSD fits within institutional payment infrastructure and settlement workflows
- Identify key stakeholders and roles in stablecoin-based payment ecosystems
- Compare traditional cross-border payment methods with stablecoin-enabled alternatives

Instructor Note

This guide contains additional teaching notes, facilitation tips, answer keys, and time estimates marked with the icon. These sections are not included in the student edition.

Module Overview

This foundational module introduces RLUSD as a regulated stablecoin designed for institutional payments, positioning it within existing payment infrastructure and settlement frameworks. Students will understand the stakeholder ecosystem, regulatory foundations, and competitive advantages over traditional methods. Total estimated delivery time: 90-110 minutes (including lab and discussion).

RLUSD Fundamentals and the Stablecoin Landscape

Estimated Time: 18 minutes

Understanding RLUSD in the Stablecoin Ecosystem

RLUSD is Ripple's enterprise-grade stablecoin, designed specifically for institutional payment infrastructure. Unlike volatile cryptocurrencies, RLUSD maintains a 1:1 peg with the US dollar through full reserve backing, combining the stability of traditional fiat with the efficiency of blockchain settlement.

Key Characteristics of RLUSD:

- **Full Reserve Backing:** Each RLUSD token is backed by US dollar deposits, short-term US government treasuries, and cash equivalents held in segregated accounts
- **Regulatory Compliance:** Issued under regulatory frameworks including New York Department of Financial Services (NYDFS) oversight
- **Multi-Ledger Support:** Native issuance on both XRP Ledger and Ethereum, enabling broad institutional integration
- **Institutional Focus:** Purpose-built for treasury operations, cross-border payments, and settlement workflows

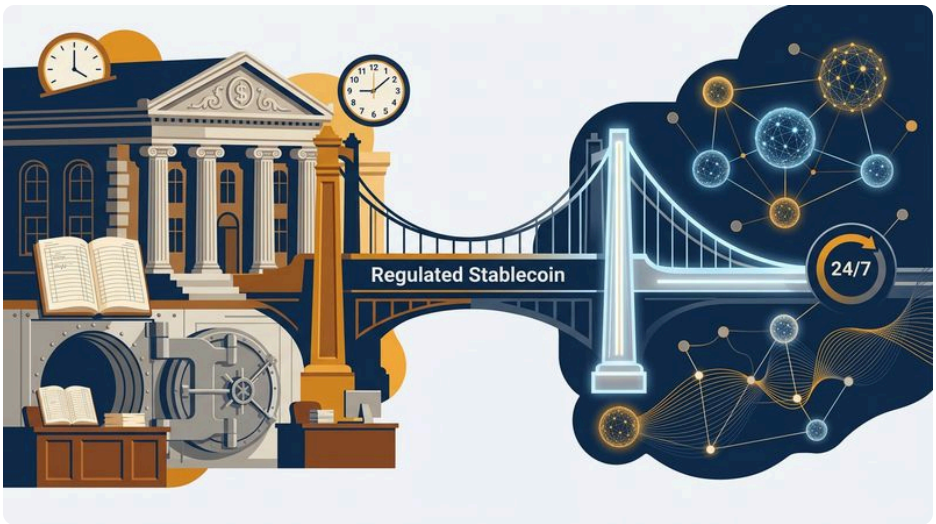
The Institutional Imperative

Traditional stablecoins like USDT and USDC pioneered digital dollar equivalents, but RLUSD differentiates itself through enterprise-specific design. Where consumer-focused stablecoins prioritize trading and DeFi integration, RLUSD emphasizes regulatory clarity, enterprise custody solutions, and seamless integration with existing payment infrastructure.

Financial institutions face mounting pressure to modernize payment rails while maintaining compliance and risk management standards. RLUSD addresses this by providing a regulated on-ramp to blockchain-based settlement without requiring institutions to navigate the complexities of unregulated digital assets.

Regulatory positioning matters: RLUSD's compliance framework enables banks, payment service providers, and corporate treasuries to adopt blockchain settlement within their existing regulatory

perimeters, reducing legal uncertainty and accelerating institutional adoption.



Abstract illustration showing digital and traditional financial infrastructure converging, with regulated stablecoin bridge connecting legacy banking systems to modern blockchain settlement

Key Takeaways

- RLUSD is a fully-reserved, regulated stablecoin designed for institutional payment infrastructure
- Regulatory compliance and multi-ledger support differentiate RLUSD from consumer-focused stablecoins
- Enterprise treasury and cross-border settlement are primary institutional use cases

Instructor Notes

Talking Points:

- RLUSD is a USD-collateralized stablecoin issued by Ripple, designed specifically for institutional use—not retail consumers
- Distinguish RLUSD from other stablecoins by emphasizing its regulatory clarity and institutional backing
- Explain the 1:1 USD backing mechanism and how this differs from algorithmic or partially-collateralized alternatives
- Position RLUSD within the broader stablecoin landscape, noting the competitive alternatives (USDC, USDT, others) and their trade-offs
- Highlight that RLUSD operates on XRP Ledger, not Ethereum, which has implications for settlement speed and infrastructure

Demo Suggestion: Display a visual comparison chart showing RLUSD vs. USDC vs. USDT characteristics (collateralization type, issuer, blockchain, regulatory status, settlement speed). Walk through a sample RLUSD transaction on the XRP Ledger to show real-time settlement in action.

Common Student Questions:

Q: Is RLUSD the same as XRP cryptocurrency?
A: No. RLUSD is a stablecoin representing USD value, while XRP is a volatile digital asset. RLUSD is issued by Ripple and collateralized 1:1 with USD. Both operate on XRP Ledger, but serve different purposes.

Q: Who can hold and use RLUSD?
A: RLUSD is designed for institutional players: banks, payment processors, and financial institutions. Retail users may access it indirectly through institutional platforms, but direct issuance focuses on regulated entities.

Q: What happens to RLUSD if the issuer faces problems?
A: RLUSD is fully collateralized with USD held in escrow accounts with regulated custodians, providing strong protection. The regulatory framework requires transparency about collateral holdings.

Integration Points with Existing Payment Infrastructure

Estimated Time: 16 minutes

Bridging Legacy and Modern Payment Systems

Institutional payment infrastructure consists of multiple interconnected layers: core banking systems, payment gateways, correspondent banking networks, and settlement platforms. RLUSD integration occurs at strategic points within this ecosystem, enabling institutions to adopt blockchain settlement without wholesale replacement of existing systems.

Primary Integration Touchpoints:

1. **Treasury Management Systems (TMS):** RLUSD wallets integrate with corporate TMS platforms, allowing treasury teams to manage stablecoin balances alongside traditional currency accounts
2. **Payment Gateways:** API-based integration enables payment processors to offer RLUSD as a settlement option within existing merchant and B2B payment flows
3. **Custody Solutions:** Institutional-grade custody providers support RLUSD, enabling secure storage with familiar risk controls and insurance frameworks
4. **Liquidity Pools:** Integration with Ripple's liquidity network allows institutions to convert between RLUSD and local currencies on-demand

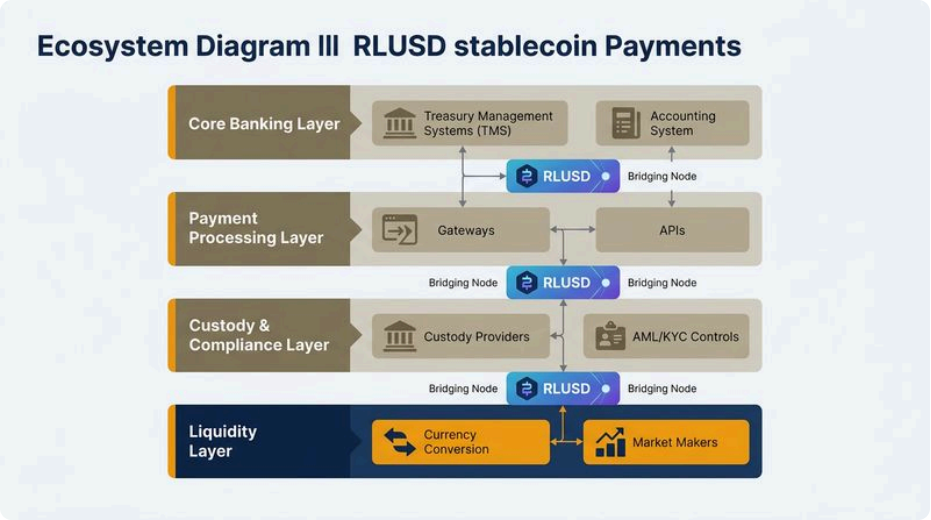
Technical Architecture Considerations

Successful integration requires understanding how blockchain settlement interacts with traditional payment messaging. While legacy systems use SWIFT, ISO 20022, and proprietary APIs, RLUSD transactions settle on-ledger through smart contracts and cryptographic signatures.

The integration challenge: Institutions must map traditional payment instructions to blockchain transactions while maintaining audit trails, compliance controls, and operational workflows. Middleware solutions, often provided by Ripple or third-party vendors, translate between these paradigms.

API-first design enables developers to abstract blockchain complexity. Rather than requiring payment operations teams to understand distributed ledgers, integration platforms present familiar

RESTful APIs that trigger RLUSD transactions behind the scenes. This approach accelerates adoption by minimizing retraining and process redesign.



Illustrated ecosystem map showing institutional payment infrastructure layers with RLUSD integration points highlighted across treasury systems, payment gateways, custody providers, and liquidity networks

Key Takeaways

- RLUSD integrates at strategic points within existing payment infrastructure rather than replacing systems
- Treasury management, payment gateways, custody, and liquidity conversion are primary integration touchpoints
- Middleware solutions translate between traditional payment messaging and blockchain settlement

Instructor Notes

Talking Points:

- RLUSD integrates with legacy banking infrastructure through Payment Service Providers (PSPs) and API-based on/off ramps
- Explain the on-ramp process (fiat RLUSD) and off-ramp process (RLUSD fiat) as critical integration points
- Describe how RLUSD coexists with SWIFT, ACH, and other legacy systems—it doesn't replace them immediately but offers a parallel path
- Emphasize that institutional adoption requires integration with existing core banking systems, treasury platforms, and compliance monitoring tools
- Discuss the role of node operators and validators in institutional settings and how they maintain access to the network

Demo Suggestion: Show a visual workflow diagram of how RLUSD flows through a bank's operations: customer deposit fiat conversion to RLUSD cross-border transfer fiat conversion on receiving end. Demonstrate how an API call initiates a transfer and how settlement confirmation appears on distributed ledger.

Common Student Questions:

Q: Do banks need to completely replace SWIFT with RLUSD?
A: No. RLUSD provides an alternative rail for specific use cases (fast, transparent, low-cost transfers). Banks will likely use both systems: SWIFT for certain corridors and relationships, RLUSD for corridors where the benefits justify implementation.

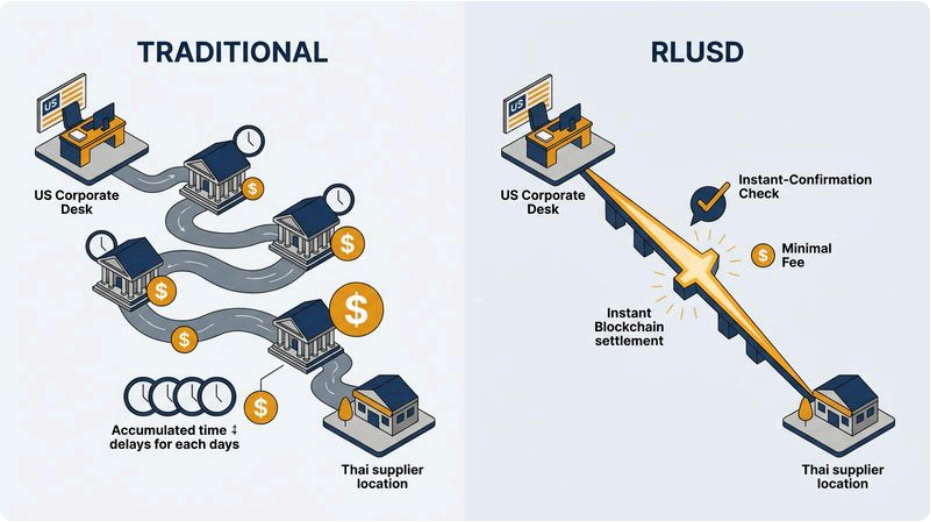
Q: What compliance systems must a bank integrate with RLUSD?
A: Banks must integrate AML/KYC screening, sanctions monitoring, and transaction reporting systems. RLUSD transfers are immutable and trackable, making compliance easier in some ways but requiring robust identity verification on-ramps.

Q: Can RLUSD work with existing payment gateways?
A: Yes, but gateways need to be enhanced. Specialized fintech partners and PSPs are building RLUSD-enabled interfaces that connect to legacy banking APIs and gateways.

Settlement Models: Real-Time Gross Settlement vs. Correspondent Banking

Estimated Time: 20 minutes

Net Settlement: Transactions batch by counterparty and settle periodically (end-of-day or longer), rather than individually and immediately



Before-and-after comparison showing traditional correspondent banking payment flow with multiple intermediary banks on left side versus direct RLUSD blockchain settlement on right side

Key Takeaways

- Correspondent banking relies on multi-hop intermediaries with batched net settlement and opaque fees
- RLUSD enables Real-Time Gross Settlement with atomic finality and transparent costs
- Blockchain-based RTGS operates 24/7/365 and eliminates settlement risk inherent in traditional multi-day clearing

Instructor Notes

Talking Points:

- Traditional correspondent banking uses a multi-hop settlement model with intermediaries, each taking time and fees (T+1 or T+2 settlement cycles)
- Real-Time Gross Settlement (RTGS) with RLUSD means immediate, final settlement without intermediaries—critical for institutions that need liquidity certainty
- Explain gross settlement (each transaction settled individually) vs. net settlement (offsetting debits and credits)—RLUSD operates as gross settlement on distributed ledger
- Highlight the liquidity advantage: with RTGS, a bank's cash position is updated immediately, improving treasury management and reducing counterparty risk
- Discuss how RLUSD enables multi-currency settlement corridors through bridge assets and atomic swaps, contrasting with correspondent banking's sequential hops

Demo Suggestion: Create two parallel timelines on whiteboard/slide: (1) Correspondent banking: Day 1 morning—request sent, Day 1 afternoon—intermediary receives and processes, Day 1 evening—receives confirmation, Day 2—funds appear. (2) RLUSD: Same time—funds transferred and settled in 4-5 seconds. Show the XRP Ledger's 4-second close interval and 99.99% uptime metrics.

Common Student Questions:

Q: Does real-time settlement eliminate counterparty risk?
A: RTGS significantly reduces counterparty risk because settlement is atomic (all-or-nothing) and immediate. However, operational risks (key management, network access) remain. It's not zero-risk but substantially better than correspondent banking's multi-day exposure.

Q: How does RLUSD handle settlement if there's a network outage?
A: XRP Ledger has 99.99% uptime and multiple redundancy across global validators. If an outage occurs, transactions queue and settle once the network recovers, but operational risk is minimized through the distributed validator model.

Q: Can RLUSD be used for batched or net settlement?
A: Technically, yes—institutions could batch transactions and use RLUSD for periodic net settlement. But the real advantage is RTGS; batching reduces that benefit. Most use cases favor real-time settlement.

Institutional Use Cases and Stakeholder Ecosystem

Estimated Time: 17 minutes

Primary Institutional Use Cases for RLUSD

Stablecoin adoption in institutional settings clusters around specific high-value use cases where traditional payment methods demonstrate clear inefficiencies. RLUSD addresses these gaps through programmable, instant settlement with regulatory compliance.

Cross-Border Treasury Operations:

Multinational corporations manage liquidity across subsidiaries in different jurisdictions. RLUSD enables treasury teams to rebalance funds across geographies instantly, reducing FX conversion costs and eliminating multi-day settlement delays that tie up working capital.

B2B Payment Settlement:

Supply chain payments, especially international supplier settlements, benefit from RLUSD's speed and cost structure. A manufacturer paying component suppliers across Asia can settle in RLUSD, with recipients converting to local currency through Ripple's liquidity network—completing in minutes what traditionally requires days.

Remittance Corridor Optimization:

Payment service providers and money transfer operators use RLUSD as a bridge asset for high-volume remittance corridors. Rather than pre-funding nostro accounts in dozens of currencies, providers hold RLUSD and convert on-demand, dramatically improving capital efficiency.

Securities Settlement:

Institutional investors and broker-dealers explore RLUSD for cash leg settlement in securities transactions. Delivery-versus-payment (DvP) workflows benefit from atomic settlement, reducing counterparty risk in securities trades.

Stakeholder Ecosystem

Successful RLUSD adoption requires coordination across a diverse stakeholder ecosystem:

- **Issuers (Ripple):** Maintain reserve backing, regulatory compliance, and technical infrastructure
- **Regulated Financial Institutions:** Banks, broker-dealers, and payment processors that custody RLUSD and facilitate conversion
- **Corporate Treasuries:** End-users managing institutional payment and liquidity operations
- **Technology Providers:** Custody platforms, payment gateways, and integration middleware vendors
- **Regulators:** Oversight bodies ensuring compliance with anti-money laundering (AML), sanctions, and capital requirements
- **Liquidity Providers:** Market makers facilitating RLUSD to fiat currency conversion

Each stakeholder contributes distinct capabilities and faces unique regulatory and operational requirements. Ecosystem health depends on clear roles, interoperable standards, and aligned incentives across participants.

Illustrated stakeholder ecosystem map showing interconnected roles including Ripple issuer, banks, corporate treasuries, technology providers, regulators, and liquidity providers arranged around RLUSD at center

Key Takeaways

- Primary institutional use cases include cross-border treasury, B2B payments, remittance optimization, and securities settlement
- RLUSD ecosystem includes issuers, financial institutions, corporate treasuries, technology providers, regulators, and liquidity providers
- Use case viability depends on stakeholder coordination and aligned incentives across the ecosystem

Instructor Notes

Talking Points:

- Primary use cases: cross-border payments (B2B remittances), liquidity management for correspondent banks, settlement of capital markets trades, and forex operations
- Key stakeholders: originating banks, receiving banks, payment service providers, fintech platforms, market makers (for RLUSD/fiat conversion), compliance vendors, and custodians
- Each stakeholder has a different motivation: banks seek cost reduction and speed, fintech platforms seek competitive advantage, regulators seek transparency and AML compliance
- Explain the role of market makers in providing liquidity and tight bid-ask spreads for RLUSD/fiat conversions—essential for adoption
- Discuss ecosystem maturity: early-stage but growing, with partnerships between banks (SBI Group, Standard Chartered, etc.) and Ripple driving adoption

Demo Suggestion: Create a stakeholder ecosystem diagram showing flow of value and information. Animate the diagram to show a specific use case: Bank A (correspondent) initiates payment, PSP converts to RLUSD, payment moves via distributed ledger, Bank B's PSP converts back to fiat, Bank B receives settlement notification and reconciles. Highlight each stakeholder's role.

Common Student Questions:

Q: Who makes money in the RLUSD ecosystem if settlement is fast and cheap?
A: Value shifts: on-ramp/off-ramp spreads (market makers), PSP service fees, compliance/monitoring tools, and APIs generate revenue. Banks save money on settlement costs and reduce working capital tied up in correspondent relationships. It's a reallocation, not a zero-sum reduction.

Q: Do I need to partner with Ripple to use RLUSD?
A: No direct partnership required. You can access RLUSD through regulated PSPs and exchanges. However, larger banks may work with Ripple for integration, strategic deployment, or gaining validator status.

Q: What happens if a market maker exits and liquidity dries up?
A: This is a real risk in early-stage ecosystems. Mitigation: multiple market makers, institutional anchors (large banks), and growing on/off-ramp infrastructure. Regulatory clarity also encourages more market makers to enter.

Value Proposition: Speed, Cost, and Transparency

> **Estimated Time:** 15 minutes

Quantifying the Institutional Value Proposition

RLUSD's value to institutional adopters manifests across three primary dimensions: settlement speed, transaction cost, and operational transparency. Understanding these benefits in quantitative terms enables ROI analysis and adoption decision-making.

Settlement Speed:

Traditional cross-border payments settle in 2-5 business days through correspondent banking networks. RLUSD transactions achieve finality in 3-5 seconds on-chain. For treasury operations managing billions in cross-border flows, this 10,000x speed improvement translates to:

- **Reduced float:** Funds available for deployment or investment within seconds rather than days
- **Improved cash forecasting:** Instant settlement eliminates uncertainty in receivables timing
- **Faster market response:** Treasury can react to FX opportunities or funding needs in real-time

Transaction Cost Structure:

Correspondent banking fees compound across intermediaries, typically ranging from 3-7% of transaction value for cross-border payments when including FX spreads and intermediary charges. RLUSD on-chain transaction fees measure in cents, with FX conversion occurring through competitive liquidity pools.

For high-volume payment corridors, cost savings can reach 40-60% compared to traditional rails. A multinational corporation processing $500 million annually in cross-border supplier payments might reduce payment costs by $15-25 million through RLUSD adoption.

Operational Transparency:

Blockchain-based settlement provides complete transaction transparency and auditability. Every RLUSD transfer is permanently recorded on-ledger with cryptographic proof, enabling:

- **Real-time reconciliation:** Automated matching of payments to invoices without manual reconciliation

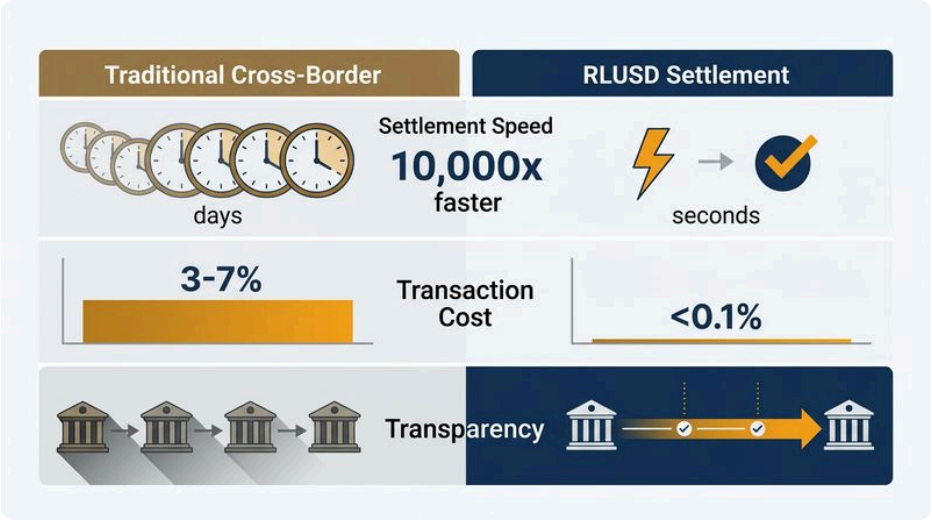
- **Audit trail compliance:** Immutable transaction history simplifies regulatory reporting and audits
- **Fraud reduction:** Transparent flows and cryptographic authentication reduce payment fraud risk

Transparency extends to fee structures—institutions know exact transaction costs upfront, eliminating hidden intermediary fees common in correspondent banking. This predictability improves financial planning and cost management.

Strategic Considerations

While speed, cost, and transparency drive initial adoption interest, strategic value emerges from competitive positioning. Early institutional adopters gain operational advantages that compound over time: stronger supplier relationships through faster payments, improved capital efficiency enabling better investment returns, and enhanced compliance positioning reducing regulatory risk.

The value proposition extends beyond direct cost savings to strategic differentiation in an increasingly digital financial ecosystem.



Infographic comparing traditional cross-border payment metrics (speed, cost, transparency) side-by-side with RLUSD stablecoin settlement metrics, showing dramatic improvements across all three dimensions

Key Takeaways

- RLUSD delivers 10,000x speed improvement over traditional cross-border settlement (seconds vs. days)
- Transaction costs can be reduced by 40-60% compared to correspondent banking in high-volume corridors
- Blockchain transparency enables real-time reconciliation, simplified audits, and reduced fraud risk
- Strategic value includes competitive positioning and operational advantages beyond direct cost savings

Instructor Notes

Talking Points:

- Speed: RLUSD transfers settle in 4-5 seconds (XRP Ledger close interval) vs. 2-3 days for correspondent banking. For institutions with global operations, this dramatically improves cash flow predictability
- Cost: Eliminate intermediate hops and associated fees (correspondent bank charges, hidden FX markups). Typical savings: 50-90% reduction in cross-border payment costs for eligible corridors
- Transparency: Immutable ledger means full visibility into transaction status, timing, and fees from initiation to settlement. Reduces reconciliation effort and fraud risk
- Emphasize that these benefits compound: faster settlement less working capital needed capital redeployment for growth; lower costs better margins or competitive pricing; transparency easier compliance audits
- Caution: Not all corridors benefit equally. Complex correspondent relationships, regulatory barriers, or low-volume pairs may not justify on-boarding costs initially

Demo Suggestion: Show a cost-benefit comparison spreadsheet: Traditional correspondent payment (e.g., USD to SGD): 3-day settlement, 4-5 fees (originating bank, correspondent, receiving bank, hidden FX markup), total cost 0.5-1.5% of amount. RLUSD payment: 4-second settlement, 2 fees (on-ramp, off-ramp), total cost 0.1-0.3%. Plot this across 1,000 annual transactions for a typical mid-size bank to illustrate annual savings.

Common Student Questions:

Q: If costs are lower, aren't banks worse off? Won't RLUSD adoption hurt bank profits?
A: Short-term, correspondent banking fees may decrease for adopting banks. But they benefit from reduced settlement risk, faster cash availability, and ability to offer better rates to customers—gaining market share. It's competitive pressure that drives ecosystem-wide efficiency.

Q: Can fees be even lower than current RLUSD pricing?
A: Potentially, as volume grows and automation increases. However, on-ramp/off-ramp spreads (market maker margin) and compliance costs are real. Natural floor exists, but it's still significantly lower than correspondent banking.

Q: Is the transparency a privacy concern?
A: RLUSD transactions are pseudonymous (address-based) not anonymous. Banks and

> institutions have full visibility for their own transactions, but the ledger doesn't expose customers' identities publicly—addresses are institution-controlled.

Regulatory Framework and Compliance Foundations

Estimated Time: 14 minutes

Regulatory Positioning of RLUSD

Stablecoin regulation represents a rapidly evolving landscape, with jurisdictions worldwide developing frameworks to govern digital asset issuance, custody, and transfer. RLUSD's regulatory strategy emphasizes proactive compliance with existing financial regulations rather than awaiting stablecoin-specific rules.

Key Regulatory Touchpoints:

- **Money Transmitter Licensing:** RLUSD issuance and redemption operates under state and federal money transmission frameworks in the US, including NYDFS oversight
- **Reserve Requirements:** Full backing with US dollar deposits and short-term treasuries aligns with proposed stablecoin reserve standards
- **AML/KYC Compliance:** Know Your Customer (KYC) and Anti-Money Laundering (AML) controls apply at issuance, redemption, and institutional integration points
- **Securities Law Considerations:** RLUSD's structure as a payment instrument rather than investment contract positions it outside securities regulation in most jurisdictions

Institutional Compliance Obligations

Financial institutions integrating RLUSD must navigate compliance requirements that extend beyond the stablecoin itself. Banks, payment processors, and broker-dealers face distinct regulatory obligations when handling digital assets:

Bank Integration Requirements:

Banks offering RLUSD custody or conversion services must address capital adequacy, liquidity coverage, and risk management frameworks. Regulatory guidance varies by jurisdiction—US banks follow OCC and Federal Reserve guidance on digital asset activities, while EU institutions navigate MiCA (Markets in Crypto-Assets) regulation.

Payment Service Provider Obligations:

PSPs incorporating RLUSD into payment flows must maintain existing AML/KYC standards, transaction monitoring, and sanctions screening. The blockchain's transparency can enhance compliance monitoring, but institutions retain responsibility for customer due diligence.

Cross-Border Regulatory Harmonization:

Institutional RLUSD adoption accelerates as regulatory frameworks harmonize across jurisdictions. Industry consortia and standard-setting bodies work to establish interoperable compliance frameworks, enabling RLUSD to flow across borders while meeting local regulatory requirements.

Understanding the regulatory foundation is essential for institutions evaluating RLUSD adoption. Compliance readiness determines timeline and operational model for integration.

Illustrated concept showing regulatory framework layers surrounding RLUSD, depicting compliance touchpoints including money transmitter licensing, reserve requirements, AML/KYC controls, and cross-border regulatory coordination

Key Takeaways

- RLUSD operates under existing money transmitter, reserve, and AML/KYC regulatory frameworks
- Institutional adopters face jurisdiction-specific compliance obligations for digital asset integration
- Regulatory harmonization across jurisdictions accelerates institutional stablecoin adoption
- Compliance readiness is a critical factor in institutional RLUSD integration timelines

Instructor Notes

Talking Points:

- RLUSD operates within a regulated framework: Ripple holds money transmitter licenses in multiple jurisdictions, and RLUSD is issued under compliance supervision
- Key regulatory requirements: Know Your Customer (KYC), Anti-Money Laundering (AML), sanctions screening, and transaction reporting. RLUSD architecture supports these through immutable audit trails
- Different jurisdictions have different stablecoin regulations: US (state money transmitter licenses + FinCEN guidance), EU (Markets in Crypto-Assets Regulation/MiCA), Singapore (Payment Services Act), Japan (Payment Services Act), etc.
- Emphasize that RLUSD's 1:1 USD collateralization and reserve requirements align with emerging global stablecoin standards, reducing regulatory arbitrage risks
- Discuss the tension between innovation (distributed ledger speed) and oversight (regulators' need to monitor and control financial system flows). Compliance tools and transparency of ledger help bridge this

Demo Suggestion: Show a simplified regulatory timeline: 2022—RLUSD announced, 2023—early deployments in Asia, 2024—expansion into more jurisdictions and asset classes. Overlay regulatory milestones (e.g., MiCA implementation, FinCEN guidance updates). Highlight which institutions are live and in which regions.

Common Student Questions:

Q: Is RLUSD compliant with all global regulations?
A: RLUSD is compliant in jurisdictions where it's deployed (Singapore, Japan, select others). It's not available everywhere yet—some jurisdictions have restrictions on stablecoins or crypto. Banks must verify RLUSD's legal status in their operating regions before adoption.

Q: What if a regulator changes rules and restricts RLUSD?
A: Regulatory risk exists, especially in early stages. However, RLUSD's focus on institutional use and full collateralization makes it lower-risk than many crypto assets. Ripple's proactive regulatory engagement and licensed operations reduce this risk compared to unregulated alternatives.

Q: How do AML/compliance teams monitor RLUSD transfers?
A: RLUSD transfers leave an immutable record on the ledger. Compliance teams use APIs to query transaction details, cross-reference with their customer records, and flag suspicious

patterns. The ledger's transparency actually simplifies AML monitoring compared to correspondent banking's opaque hops.

Q: Do stablecoin transfers escape capital controls or sanctions?
A: No. RLUSD's on/off-ramp requirements (fiat conversion) are the control points—same as traditional banking. Institutions must comply with sanctions and capital controls. The distributed ledger doesn't bypass these; it just provides a faster intermediate settlement layer.

Lab: Exploring RLUSD Characteristics and Payment Infrastructure Integration

Objective: Navigate Ripple's official documentation and tools to investigate RLUSD characteristics, regulatory framework, and integration points within institutional payment infrastructure, then compare traditional payment methods with stablecoin-enabled alternatives

Estimated Time: 45 minutes

Platform: Windows PC with web browser and internet access

Prerequisites:

- Windows 10 or 11 computer
- Modern web browser (Chrome, Edge, or Firefox)
- Internet connection
- Basic familiarity with web navigation
- Notepad or text editor for recording observations

Lab Facilitation

Setup: Pre-lab: Ensure all students have access to a sandboxed XRP Ledger explorer (e.g., xrpscan.com or a local testnet instance). Create test accounts with a small amount of XRP and RLUSD for each student (or provide credentials to shared accounts). Prepare a comparison worksheet with blank fields for students to fill in. Set up a demo wallet that shows a live RLUSD transfer during the walkthrough. Have backup documentation (API specs, account creation guides) available. If using testnet, verify that faucets are functional and test the explorer interface beforehand.

Pacing: Allocate 45-60 minutes for this lab. Start with 10 minutes of guided walkthrough: show how to access the explorer, what fields to look for in a transaction, and how to interpret settlement details. Then, break students into pairs or small groups and assign specific tasks: (1) locate a live RLUSD transaction and document its settlement time, (2) compare it with a sample SWIFT transfer scenario (hypothetical timeline), (3) identify fees and parties involved. Rotate between groups, checking understanding and answering questions. Finish with 5-10 minutes of group debrief: ask each group to share one finding. Don't rush—conceptual clarity is more important than completing all tasks.

Procedures

Step 1: Open your web browser and navigate to the official Ripple RLUSD documentation page

Launch your preferred web browser (Chrome, Edge, or Firefox) and enter the following URL in the address bar:

```
https://ripple.com/solutions/crypto/rlusd/
```

Press **Enter** to load the page. This is Ripple's primary information page for RLUSD stablecoin.

Expected Output:
The Ripple RLUSD landing page loads, displaying an overview of the RLUSD stablecoin, its regulatory framework, and key features. The page header should show the Ripple logo and navigation menu.

Verification: Confirm you see content describing RLUSD as a USD-backed stablecoin and references to regulatory compliance.

Step 2: Locate and document the fundamental characteristics of RLUSD

Scroll through the RLUSD page and identify the following fundamental characteristics. Open Notepad (press **Windows Key + R**, type `notepad`, press **Enter**) and create a document to record your findings:

1. What currency is RLUSD pegged to?
2. What is the backing mechanism (what assets back RLUSD)?
3. Which blockchain(s) support RLUSD?
4. What regulatory framework is mentioned?

Format your notes as:

```
RLUSD Fundamental Characteristics
==================================
Pegged Currency: [your answer]
Backing Mechanism: [your answer]
Blockchain Support: [your answer]
Regulatory Framework: [your answer]
```

> **Expected Output:**
> A Notepad document with documented characteristics showing RLUSD is pegged 1:1 to USD, backed by US dollar deposits and short-term treasuries, supported on XRP Ledger and Ethereum, and compliant with regulatory frameworks including New York Department of Financial Services oversight.

Verification: Review your notes to ensure all four characteristics are documented with specific details from the Ripple website.

Step 3: Navigate to the RLUSD technical documentation

In your browser, navigate to the Ripple developer documentation for RLUSD:

```
https://docs.ripple.com/
```

Once the page loads, use the search function (typically **Ctrl + F** or look for a search box in the navigation) and search for:

```
RLUSD
```

Click on the most relevant result related to RLUSD stablecoin documentation or integration guides.

> **Expected Output:**
> The Ripple documentation site loads, and search results display links to RLUSD-related technical documentation, integration guides, or API references.

Verification: Confirm you have accessed technical documentation that includes integration details, not just marketing content.

Step 4: Identify integration points with existing payment infrastructure

Within the technical documentation, locate sections describing how RLUSD integrates with existing payment systems. Look for:

- API endpoints or integration methods
- Settlement workflow descriptions
- Treasury management interfaces
- Bank account connection requirements

Add a new section to your Notepad document:

```
RLUSD Integration Points
========================
API Integration: [describe available APIs]
Settlement Process: [describe the workflow]
Treasury Management: [describe treasury interfaces]
Traditional Banking Links: [describe bank connections]
```

> **Expected Output:**
> Documentation showing Ripple Payments API endpoints, webhook integrations for settlement notifications, treasury dashboards for liquidity management, and requirements for fiat on/off ramps through banking partners.

Verification: Your notes should include at least three distinct integration points with specific technical details.

Step 5: Access the XRPL Learning Portal to understand blockchain settlement

Open a new browser tab and navigate to the XRP Ledger Learning Portal:

```
https://learn.xrpl.org/
```

From the main page, locate and click on the **Course** or **Lessons** section. Look for content related to:

- Payment settlement on XRPL
- How transactions are validated
- Settlement finality timing

Click on the most relevant introductory lesson about XRPL payments or settlement.

> **Expected Output:**
> The XRPL Learning Portal displays an organized curriculum with lessons covering XRPL fundamentals, payment mechanics, and settlement characteristics. You should see interactive content or videos explaining how the ledger processes payments.

Verification: Confirm you have accessed educational content that explains the XRPL transaction validation and settlement process.

Step 6: Document real-time gross settlement (RTGS) characteristics

Within the XRPL learning content, identify information about settlement timing and finality. Pay particular attention to:

- How long transactions take to settle
- Whether settlement is final or reversible
- How this compares to traditional payment rails

Add this section to your Notepad document:

```
RTGS Characteristics via XRPL
==============================
Settlement Time: [typical time for finality]
Reversibility: [can transactions be reversed?]
Validation Method: [how are transactions confirmed?]
Comparison to Traditional: [key differences]
```

Expected Output:
Notes showing XRPL transactions settle in 3-5 seconds, settlement is final and irreversible once validated by consensus, transactions are validated through the XRPL consensus protocol, and this is significantly faster than traditional correspondent banking which takes 1-5 business days.

Verification: Your notes should clearly distinguish between XRPL settlement speed and traditional payment rails.

Step 7: Navigate to Ripple's API documentation with interactive tools

Open a new browser tab and navigate to Ripple's API reference:

```
https://docs.ripple.com/api/
```

Or search for "Ripple Payments API" from the docs.ripple.com homepage.

Look for API endpoints related to payment operations. Common endpoints might include:

- `/v1/payments` - for initiating payments
- `/v1/quotes` - for obtaining payment quotes
- `/v1/accounts` - for account management

Click on any endpoint that has a **Try it** or **Test** button.

Expected Output:
The Ripple API documentation loads with a list of REST API endpoints. Each endpoint should show HTTP methods (GET, POST), parameters, and potentially an interactive testing interface.

Verification: Confirm you can see API endpoint documentation with parameter descriptions and example requests/responses.

Step 8: Explore an interactive API example for payment quotes

Locate the quotes or payment estimation endpoint (often `/v1/quotes` or similar). If an interactive "Try it" feature is available:

1. Click the **Try it** or **Test this endpoint** button
2. Examine the required parameters (source currency, destination currency, amount)
3. Review example values or documentation
4. Note the response structure showing settlement details

If no interactive feature is available, carefully read the example request and response JSON.

Add to your Notepad document:

```
Payment Quote API Observations
==============================
Endpoint: [API path]
Required Parameters: [list parameters]
Response Data: [what information is returned]
Settlement Information: [how settlement details are provided]
```

> **Expected Output:**
> Documentation showing a quote endpoint that accepts source amount, source currency (USD), destination currency, and returns estimated fees, exchange rates, settlement time estimates, and liquidity source information.

Verification: Your notes should include the specific API endpoint path and at least three parameters or response fields.

Step 9: Compare traditional correspondent banking with RLUSD settlement

Using information gathered from previous steps and additional browsing of Ripple's use cases or solutions pages, create a comparison table in your Notepad document:

```
Traditional vs. RLUSD Settlement Comparison
==========================================

Criteria            | Traditional Banking | RLUSD/XRPL
-------------------------------------------------------
Settlement Time      | [fill in]          | [fill in]
Operating Hours      | [fill in]          | [fill in]
Intermediary Banks   | [fill in]          | [fill in]
Fee Structure        | [fill in]          | [fill in]
Transparency         | [fill in]          | [fill in]
Currency Conversion  | [fill in]          | [fill in]
```

Fill in each cell based on your research. If needed, navigate to:

```
https://ripple.com/use-cases/
```

to find comparative information about traditional vs. blockchain-based payments.

> **Expected Output:**
> A completed comparison table showing: Traditional (1-5 days, business hours only, multiple intermediaries, 3-7% fees, limited transparency, multiple conversions) vs. RLUSD/XRPL (3-5 seconds, 24/7/365, direct settlement, minimal fees, full transparency, single conversion or none if USD-USD).

Verification: Your table should have all six criteria filled in with specific, contrasting details for both columns.

Step 10: Identify key stakeholders in the RLUSD ecosystem

Navigate back to the main Ripple website and explore the following sections to identify stakeholder roles:

```
https://ripple.com/customers/
https://ripple.com/partners/
```

Identify at least five distinct stakeholder categories involved in RLUSD-based payments.

Create a new section in your Notepad document:

```
Key Stakeholders in RLUSD Ecosystem
===================================
1. [Stakeholder type]: [Role/responsibility]
2. [Stakeholder type]: [Role/responsibility]
3. [Stakeholder type]: [Role/responsibility]
4. [Stakeholder type]: [Role/responsibility]
5. [Stakeholder type]: [Role/responsibility]
```

Expected Output:
A list identifying stakeholders such as: (1) Financial Institutions - issue and redeem RLUSD, (2) Payment Service Providers - facilitate transactions, (3) Enterprises - send/receive payments, (4) Liquidity Providers - ensure market depth, (5) Regulators - oversee compliance, (6) Custodians - secure digital assets, (7) Technology Partners - provide infrastructure.

Verification: You should have documented at least five distinct stakeholder types with their specific roles in the ecosystem.

Step 11: Document institutional use cases for RLUSD

Browse Ripple's use case pages and case studies to identify specific institutional applications. Look for:

```
https://ripple.com/use-cases/
```

Navigate to sections covering:

- Cross-border payments
- Treasury operations
- Trade finance
- Remittances

Add to your Notepad document:

```
Institutional Use Cases for RLUSD
==================================
Use Case 1: [Name and brief description]
Use Case 2: [Name and brief description]
Use Case 3: [Name and brief description]
Use Case 4: [Name and brief description]
```

For each use case, note the specific business problem solved and the value delivered.

Expected Output:
Documentation of four institutional use cases such as: (1) Cross-border B2B payments - reducing settlement time from days to seconds, (2) Treasury liquidity management - optimizing working capital across global subsidiaries, (3) Trade finance settlement - accelerating letter of credit processing, (4) High-volume remittances - reducing transaction costs for payment corridors.

Verification: Each use case should include both the application name and the specific business value or problem solved.

Step 12: Explore RLUSD transparency features using blockchain explorers

Navigate to the XRP Ledger explorer to understand transaction transparency:

```
https://livenet.xrpl.org/
```

Or alternatively:

```
https://xrpscan.com/
```

Examine the interface to understand what information is publicly visible:

- Transaction history
- Account balances
- Settlement timestamps
- Network validators

You don't need to search for specific transactions, just observe the available information categories.

Add to your Notepad:

```
RLUSD Transparency Capabilities
================================
Publicly Visible Information:
- [list observable data points]

Privacy Protections:
- [note what is NOT visible]
```

Expected Output:
Notes showing publicly visible information includes transaction hashes, timestamps, amounts, sender/receiver addresses (but not identities), and settlement status. Privacy protections include: actual identity of participants remains pseudonymous, business purposes not disclosed, and individual transaction details can be permissioned.

Verification: Your notes should distinguish between what blockchain transparency provides and what privacy protections exist for institutional users.

Step 13: Research regulatory compliance framework for RLUSD

Return to the Ripple RLUSD page:

```
https://ripple.com/solutions/crypto/rlusd/
```

Scroll to sections covering regulatory compliance, trust, and reserves. Look for specific mentions of:

- Regulatory authorities providing oversight
- Attestation and audit processes
- Reserve management and transparency

Document your findings:

```
RLUSD Regulatory Framework
==========================
Regulatory Oversight: [which authorities]
Attestation Process: [how reserves are verified]
Reserve Composition: [what backs each RLUSD]
Transparency Reports: [how often and what content]
```

Expected Output:
Notes indicating RLUSD is issued under oversight of the New York Department of Financial Services (NYDFS), backed 1:1 by USD deposits and short-term US Treasury securities, subject to monthly third-party attestations of reserves, with public transparency reports available.

Verification: Your notes should specifically name at least one regulatory authority and describe the reserve verification process.

Step 14: Calculate theoretical payment cost savings

Using the comparison data you've collected, calculate theoretical cost savings for a sample international payment scenario.

Open Calculator (press **Windows Key + R**, type `calc`, press **Enter**).

Scenario: A company sends $100,000 USD cross-border monthly.

Traditional cost assumptions:

- Wire transfer fee: $40
- Intermediary bank fees: $25-35 (use $30)
- FX markup: 2-3% (use 2.5%)

RLUSD cost assumptions:

- Network fee: ~$0.01
- FX markup: 0.1-0.5% (use 0.3%)

Calculate:

1. Traditional total cost = Wire fee + Intermediary + (Amount × FX markup %)
2. RLUSD total cost = Network fee + (Amount × FX markup %)
3. Savings = Traditional cost - RLUSD cost
4. Savings percentage = (Savings / Traditional cost) × 100

Document in Notepad:

```
Cost Savings Calculation
========================
Monthly Payment Amount: $100,000

Traditional Method:
Wire Fee: $40.00
Intermediary Fee: $30.00
FX Markup (2.5%): $[calculate]
Total Traditional Cost: $[calculate]

RLUSD Method:
Network Fee: $0.01
FX Markup (0.3%): $[calculate]
Total RLUSD Cost: $[calculate]

Monthly Savings: $[calculate]
Savings Percentage: [calculate]%
Annual Savings: $[monthly × 12]
```

Expected Output:
Completed calculation showing: Traditional ($40 + $30 + $2,500 = $2,570), RLUSD ($0.01 + $300 = $300.01), Monthly Savings ($2,269.99), Savings Percentage (88.3%), Annual Savings ($27,239.88).

Verification: All calculation fields should be filled with specific dollar amounts and percentages.

Step 15: Save your research findings and review learning outcomes

In Notepad, click **File > Save As**.

Navigate to your Documents folder:

1. In the Save As dialog, click **Documents** in the left sidebar
2. In the "File name" field, type:

   ```
   RLUSD_Lab_Findings.txt
   ```

3. Ensure "Save as type" is set to **Text Documents (*.txt)**
4. Click **Save**

Review your complete document and verify you have documented:

- RLUSD fundamental characteristics
- Integration points with payment infrastructure
- RTGS settlement characteristics
- API structure and functionality
- Traditional vs. RLUSD comparison
- Stakeholder ecosystem
- Institutional use cases
- Transparency capabilities
- Regulatory framework
- Cost savings analysis

Close all browser tabs and Notepad.

Expected Output:
A saved text file named "RLUSD_Lab_Findings.txt" in your Documents folder containing all research findings organized in clearly labeled sections.

Verification: Navigate to Documents folder and confirm the file exists. Open it to verify all sections are present and populated with your research findings.

Troubleshooting

Problem: Ripple website pages return 404 errors or content has moved
Solution: Navigate to https://ripple.com and use the site's main navigation menu to find RLUSD information under 'Solutions' or 'Products'. Use the search function (magnifying glass icon) and search for 'RLUSD' to locate current documentation. Ripple occasionally reorganizes their site structure.

Problem: API documentation 'Try it' features require authentication or account creation
Solution: If interactive features require login, focus on reading the example requests and responses provided in the documentation. The learning objective is understanding the API structure, not executing actual API calls. Document the example JSON structures and parameter requirements shown in the static documentation.

Problem: XRPL Learning Portal content is too technical or assumes blockchain knowledge
Solution: Focus on high-level concepts rather than technical implementation details. Look for introductory or 'Getting Started' sections. Skip sections about consensus algorithms or cryptography and focus on payment flow diagrams and settlement timing explanations. Watch any available video content which often explains concepts more clearly.

Problem: Cannot find specific cost or fee information for RLUSD transactions
Solution: RLUSD fees vary by implementation and use case. For this lab, use the provided assumptions in Step 14. In production scenarios, fees are determined by the financial institution or payment service provider implementing RLUSD, not by a fixed public rate. Focus on the comparative methodology rather than exact figures.

Problem: Blockchain explorer shows complex technical data that is difficult to interpret
Solution: Don't try to understand every field in the explorer. Focus only on these observable elements: transaction timestamps (showing settlement speed), transaction status (showing finality), and the presence of sender/receiver addresses (showing transparency). You can ignore technical fields like 'Sequence Number', 'LastLedgerSequence', or cryptographic hashes.

Common Student Errors

Students confuse XRP transactions with RLUSD transactions on the ledger.
Clarify that both are recorded on the same XRP

Challenge Exercise

Institutional Payment Corridor Analysis

Select a specific international payment corridor (e.g., USD to EUR, USD to MXN, or USD to PHP) that represents a real business need for an institution you're familiar with or interested in.

Using the research skills developed in this lab, investigate and document:

1. **Current State Analysis**: Research the traditional payment infrastructure for this corridor - identify typical settlement times, costs, intermediary banks involved, and common pain points. Use resources like SWIFT documentation, World Bank remittance data, or financial institution websites.
2. **RLUSD Integration Design**: Describe how RLUSD could be integrated into this payment corridor. Consider: What role would RLUSD play? Would it replace or complement existing rails? What stakeholders would need to participate? What regulatory considerations exist for both the source and destination jurisdictions?
3. **Value Quantification**: Create a detailed cost-benefit analysis comparing traditional methods to an RLUSD-enabled solution for a realistic transaction volume (e.g., 100 transactions per month at varying amounts). Include not just direct costs but also opportunity costs of slow settlement and working capital implications.
4. **Implementation Roadmap**: Outline a high-level implementation plan including: required integrations, stakeholder onboarding sequence, regulatory approvals needed, and a phased rollout strategy.

Deliverable: Create a 2-3 page document (you may use Word, Google Docs, or continue with formatted text) presenting your analysis with specific data points, calculations, and actionable recommendations. Include references to specific Ripple documentation, regulatory frameworks, or industry sources you consulted.

Answer Key

1. What is the primary differentiator of RLUSD compared to consumer-focused stablecoins like USDT?

A) RLUSD is designed specifically for institutional payment infrastructure with regulatory compliance emphasis

B) RLUSD maintains a higher peg ratio to the US dollar

C) RLUSD is only available on the XRP Ledger

D) RLUSD offers higher interest yields to token holders

Answer: A

RLUSD differentiates itself through enterprise-specific design emphasizing regulatory clarity, institutional custody solutions, and seamless integration with existing payment infrastructure, rather than focusing on consumer trading or DeFi use cases.

2. In traditional correspondent banking networks, what is a primary source of settlement risk?

A) Excessive transaction speed creating operational errors

B) Time lag between payment initiation and final settlement creating counterparty exposure

C) Overly transparent fee structures reducing bank profitability

D) Direct settlement between payer and payee without intermediaries

Answer: B

Correspondent banking's multi-day settlement cycles create windows of counterparty exposure between initiation and final settlement. This settlement risk is a fundamental limitation that Real-Time Gross Settlement via RLUSD addresses through near-instant finality.

3. Which institutional use case benefits most from RLUSD's ability to reduce pre-funded nostro account requirements?

A) Securities settlement for broker-dealers

B) Remittance corridor optimization for payment service providers

C) Internal treasury operations within a single jurisdiction

D) Domestic ACH payment processing

Answer: B

Payment service providers and money transfer operators traditionally pre-fund nostro accounts in multiple currencies to serve various remittance corridors—often 20-50+ currency pairs globally. RLUSD enables these providers to hold stablecoin balances as a bridge asset and convert to local currencies on-demand through liquidity networks, dramatically improving capital efficiency by eliminating the need for simultaneous pre-funding across dozens of currency pairs.

4. What is the approximate settlement time for RLUSD transactions compared to traditional cross-border payments?

A) RLUSD settles in 30 minutes compared to traditional 24-hour settlement

B) RLUSD settles in 3-5 seconds compared to traditional 2-5 business days

C) RLUSD settles in 1 hour compared to traditional same-day settlement

D) RLUSD and traditional methods have similar settlement times but different cost structures

Answer: B

RLUSD achieves transaction finality in 3-5 seconds on-chain, representing approximately a 10,000x speed improvement over traditional cross-border payments that typically settle in 2-5 business days through correspondent banking networks.

5. Which regulatory framework component is essential for RLUSD's institutional positioning?

A) Securities registration as an investment contract

B) Full reserve backing with US dollar deposits and short-term treasuries

C) Exemption from all AML/KYC requirements to enable faster transactions

D) Classification as a commodity futures contract

Answer: B

RLUSD maintains a 1:1 peg with the US dollar through full reserve backing consisting of US dollar deposits, short-term US government treasuries, and cash equivalents. This reserve structure aligns with proposed stablecoin standards and is fundamental to RLUSD's regulatory positioning and institutional credibility.

6. How does blockchain transparency benefit institutional compliance operations?

A) It eliminates the need for AML/KYC customer due diligence

B) It enables real-time reconciliation and provides immutable audit trails for regulatory reporting

C) It allows institutions to hide transaction details from regulators

D) It reduces all compliance costs to zero

Answer: B

Blockchain-based settlement provides complete transaction transparency with every RLUSD transfer permanently recorded on-ledger. This enables automated real-time reconciliation of payments to invoices and creates immutable transaction histories that simplify regulatory reporting and audits. However, institutions still retain responsibility for AML/KYC customer due diligence.

7. What is a primary integration challenge when connecting RLUSD to existing institutional payment systems?

A) RLUSD transactions are too slow for institutional requirements

B) Mapping traditional payment messaging standards to blockchain transactions while maintaining audit trails

C) RLUSD cannot integrate with treasury management systems

D) Blockchain settlement operates only during banking hours

Answer: B

Traditional payment systems use SWIFT, ISO 20022, and proprietary APIs with standardized message formats, while RLUSD settles on-ledger through smart contracts and cryptographic signatures with JSON data structures. The integration challenge involves mapping traditional payment instructions (e.g., SWIFT MT103 messages) to blockchain transactions while maintaining compliance controls, audit trails, and operational workflows that treasury teams expect. Middleware or integration platforms—such as those provided by Ripple or third-party vendors—abstract this complexity by translating between payment messaging standards and blockchain APIs.

8. For a multinational corporation processing $500 million annually in cross-border supplier payments, what is the potential cost savings range from RLUSD adoption?

A) $1-2 million (approximately 0.2-0.4% savings)

B) $5-8 million (approximately 1-1.6% savings)

C) $15-25 million (approximately 3-5% savings, representing 40-60% cost reduction)

D) $50-100 million (approximately 10-20% savings)

Answer: C

Correspondent banking fees typically range from 3-7% of transaction value when including FX spreads and intermediary charges. RLUSD can reduce costs by 40-60% in high-volume corridors. For $500 million in annual cross-border payments at traditional 3-7% costs ($15-35 million), a 40-60% reduction yields approximately $15-25 million in savings.

Module Summary

This module introduced RLUSD as a regulated, enterprise-grade stablecoin designed for institutional payment infrastructure. You explored how RLUSD integrates with existing payment systems, enabling Real-Time Gross Settlement that dramatically improves speed, cost, and transparency compared to traditional correspondent banking. The module examined institutional use cases across treasury operations, B2B payments, remittances, and securities settlement, while identifying the diverse stakeholder ecosystem required for successful adoption. Finally, you reviewed the regulatory framework positioning RLUSD within existing financial compliance requirements. Understanding these fundamentals prepares you to evaluate RLUSD's technical architecture and integration strategies in subsequent modules.